An introduction to atomic structure

All matter is built from particles called **atoms**. Atoms are the building blocks of all materials on earth. More than 100 types of atoms have been discovered. We call them **elements**.

The structure of an atom

For a long time people thought that atoms were the smallest particles in existence. We now know that atoms are made up of even smaller **sub-atomic particles**: electrons, protons and neutrons.

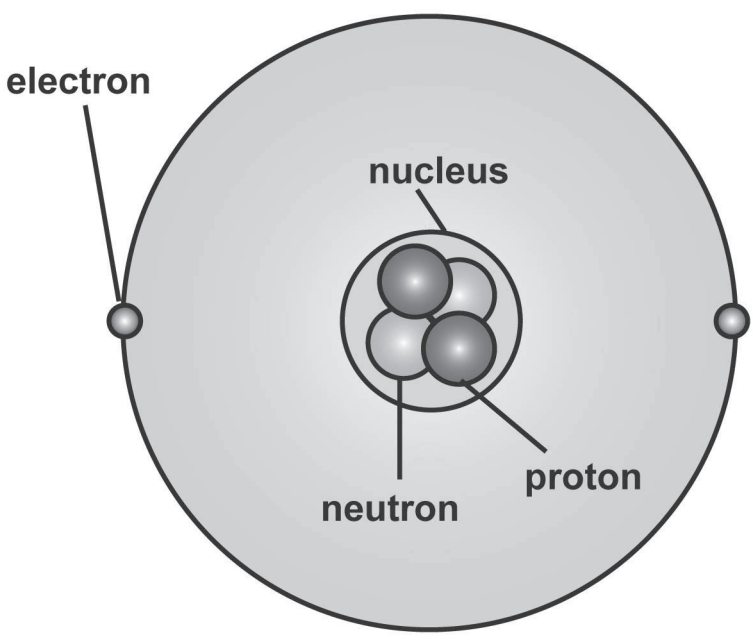
- **Electrons** are very tiny and light. They have a **negative** charge.
- **Protons** are much larger and heavier than electrons and have a **positive** charge.
- **Neutrons** are larger and heavier like protons, but have no electrical charge (they are **neutral**).

How are the particles arranged?

An atom has a **nucleus** in its centre. Protons and neutrons can be found inside the nucleus. Electrons are constantly spinning around the outside of the nucleus. Each atom has an equal number of protons and electrons. This means it has no overall **charge**.

Scientists use simple diagrams to show the structure of an atom. This is a diagram of a helium atom. It contains:

- 2 protons
- 2 neutrons
- 2 electrons.



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Atomic structure

Read the Resource sheet 'An introduction to atomic structure', then answer the following questions.

☞ **1** Use the words in the word bank to complete the following passage. Some words will be used more than once.

Word bank
three
atom
electrons
nucleus
substances
100
neutrons
spinning
one
copper
protons
unique

Elements are simple s_____ made up of the same type of a_____. There are more than 1_____ different types of elements. Each element's atom is u_____. For example, c_____ atoms are different from silver atoms.

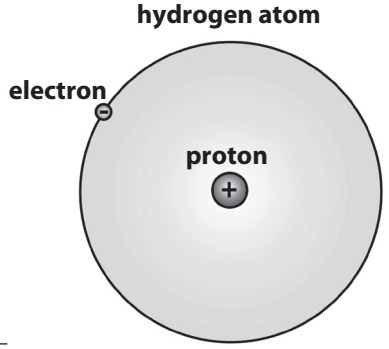
Atoms are made up of t_____ different sub-atomic particles: n_____, protons and e_____. The middle part of the atom is called the n_____. The nucleus contains p_____ and neutrons. The electrons are found s_____ around the outside of the n_____.

An atom has the same number of protons and e_____. For example, hydrogen has one proton, so it also has o_____ electron.

2 Match the sentence halves to complete the information below.

Protons and neutrons are	no charge. This means they are neutral.
Protons have a	can be found spinning around the nucleus.
Electrons are tiny and	positive charge.
Neutrons have	about the same size and mass.
Electrons have	a negative charge.

3 Look at the diagram of a hydrogen atom.



a How many protons does a hydrogen atom have? _____

b How many neutrons does a hydrogen atom have? _____

c How many electrons does a hydrogen atom have? _____

d What is unusual about hydrogen? _____

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